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Project information

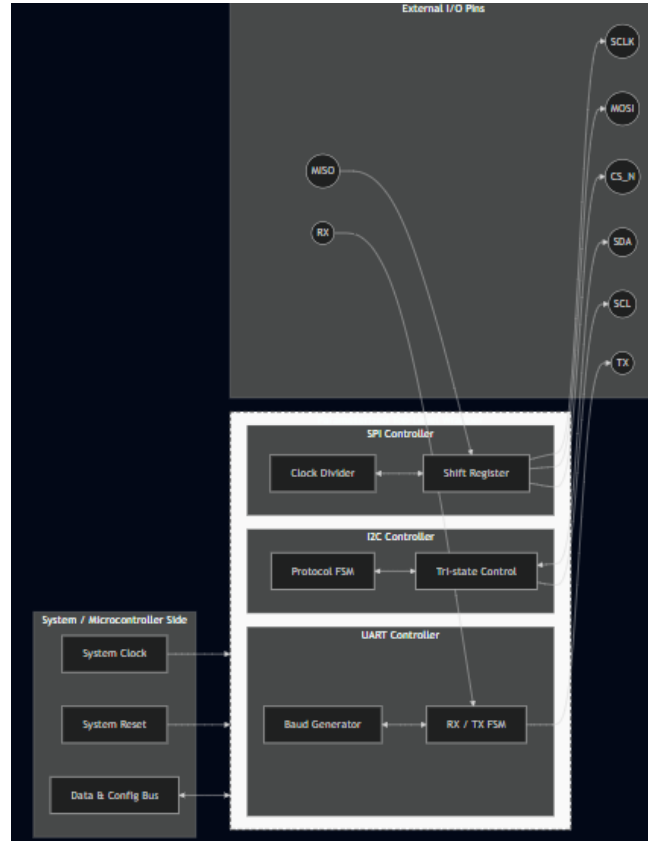
- Goal

The primary objective is to design a highly configurable, reusable Universal Serial IP block encompassing SPI, I2C, and UART protocols tailored for the GF180 process node. By utilizing an open-source RTL-to-GDSII flow, this foundational building block ensures robust and area-efficient peripheral communication for future system-on-chip integrations.

- Design - High Level Proposal

The hardware architecture features a parameterized VHDL top-level wrapper that seamlessly integrates three distinct serial communication IP blocks. Internally, the design comprises a Baud Rate Generator and RX/TX FSMs for asynchronous UART, a Shift Register and Clock Divider for full-duplex SPI, alongside a complex Protocol FSM and Tri-state Buffer Control for half-duplex I2C. These specific sub-modules will undergo rigorous functional testbench verification before being synthesized, placed, and routed into the GF180 node utilizing the LibreLane RTL-to-GDSII flow.

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- Application

This controller acts as a reliable digital bridge for embedded systems, enabling seamless data acquisition from various external peripheral modules such as industrial field sensors and debugging terminals.

Team background

Our team is composed of three third year and one second year undergraduate Electrical Engineering Students from Diponegoro University, Indonesia. **We have no prior IC design Experience.**

Questions, Suggestions, Doubts?